Gianni Antichi

Curriculum Vitae

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Employment

2018–Present **Assistant Professor**, Queen Mary University of London, London, UK.
School of Electronic Engineering and Computer Science

2018–Present **Research Fellow**, Alan Turing Institute, London, UK.

2015–2018 **Senior Research Associate**, University of Cambridge, Cambridge, UK.
Department of Computer Science and Technology

2013–2015 **Research Associate**, University of Cambridge, Cambridge, UK.
Department of Computer Science and Technology

2011–2013 **Research Associate**, University of Pisa, Pisa, IT.
Department of Information Engineering

Education

2008–2011 **PhD in Information Engineering**, *University of Pisa*, Pisa, IT.
Department of Information Engineering
Title of the dissertation: Fast Packet Processing on High Performance Architectures
Supervisor: Prof. Stefano Giordano

2005–2007 **MEng in Telecomunication Engineering**, *University of Pisa*, Pisa, IT.
Department of Information Engineering
Title of the dissertation: A Traffic Generator for Network Processors
Supervisor: Prof. Stefano Giordano

2001–2005 **BEng in Telecomunication Engineering**, *University of Pisa*, Pisa, IT.
Department of Information Engineering

Funding

2020–2023 **EPSRC New Investigator Award**, *NEAT: NEtwork dATa plane measurements as first class primitive*, (PI), 390.000 GBP.
In collaboration with VMware, Barefoot Networks, Cisco Systems and CERN

2018–2021 **Alan Turing Institute**, *LIME: Learning-based reactive Internet Engineering*, (CoI), 87.000 GBP.

2018–2020 **EPSRC Standard**, *EARL: sdn EnAbled measuRements for alL*, (CoI), 2.000.000 GBP.
A collaboration between Queen Mary University of London, University of Cambridge, LINX, ONF, Geant, Cambridge Cybercrime Centre, Corsa, Cisco, ESnet and Google

2017–2020 **NEC Research Lab Europe donation**, *High Performance Network Functions on Programmable Data Planes*, (PI), 114.000 GBP.

2015–2017 **EU H2020**, *ENDEAVOUR: Towards a flexible software-defined network ecosystem*, (CoI), 4.200.000 EUR.
A collaboration between Queen Mary University of London, University of Cambridge, Universite' Catholique de Louvain, LAAS CNRS France, IBM Research Zurich and DE-CIX

2015–2017 **EU H2020**, *SSICLOPS: Scalable and Secure Infrastructures for Cloud Operations*, (CoI), 6.200.000 EUR.
In collaboration between University of Cambridge, NetApp, Aalto University, Hasso Plattner Institut, F-Secure, Helsinki Institute of Physics, University of Bucharest, Orange Telecom, Technical University of Munich, Martel Consulting, University of Pisa, RWTH Aachen, NEC Research Europe

Teaching

2018–Present **Computer Systems and Networks**, *(ECS430U)*, Queen Mary University of London.

2018–Present **Internet Protocols and Applications**, *(ECS524U)*, Queen Mary University of London.

2018–Present **Distributed Systems**, *(ECS656U/ECS796P)*, Queen Mary University of London.

2019–2020 **21st Century Networks**, *(ECS703U/ECS703P)*, Queen Mary University of London.

2014 **Building an Internet Router**, *(P33)*, University of Cambridge.
(teaching assistant)

2010–2017 **Networking and Internet**, *Telecommunication Engineering*, University of Pisa.
(invited lecturer)

Professional Service

Editorial Board: ACM Computer Communication Review (2020–Present)

PC Chair: IEEE EuroP4 2019

Treasurer: IEEE ICNP 2018

Publication Chair: ACM HotNets 2021

Reproducibility Chair: ACM CoNEXT 2021

TPC: ACM CoNEXT 2021, IEEE ICNP 2021, TMA 2021, IEEE ICNP 2020, ACM SOSR 2020, IEEE/IFIP NOMS 2020, IEEE/ACM HPSR 2019, IEEE ICNP 2019, ACM CoNEXT 2019, ACM SOSR 2019, ACM SIGCOMM (poster session) 2018, ACM SOSR (poster session) 2018, IEEE/ACM HPSR 2017, PAM 2017

Reviewer: IEEE Journal on Selected Area of Communication, IEEE/ACM Transactions on Networking, IEEE Transactions on Network Service and Management, IEEE Network, IEEE Transactions on Parallel and Distributed Systems, IEEE Communications Surveys and Tutorials, Elsevier Computer Networks, Elsevier Computer Communications

Awards

2020 ACM SIGCOMM Student Research Competition with Sebastiano Miano: *Automatic Optimization of Software Data Planes*

2020 Facebook Networking Systems Request for Proposals: *Automatic Optimization of Software Data Planes*

2017 Best Paper award at ACM SIGCOMM with the paper: *Re-architecting datacenter networks and stacks for low latency and high performance*

2008 Best paper award at IEEE SPECTS with the paper: *BRUNO: A High Performance Traffic Generator for Network Processor*

Experiences

2016–2017 Research Associate, Open Networking Foundation (ONF)

2013 Visiting Researcher, Stanford University, US

2010 Visiting PhD student, University of Cambridge, UK

Tutorials

2017 NetFPGA camp, Telecom ParisTech, Paris, FR

2017 NetFPGA camp, University of Cambridge, Cambridge, UK

2015 Open Hardware Networking, ACM SIGCOMM, London, UK

2015 Open Source Networking, IEEE NetSoft, London, UK

2013 NetFPGA camp, Poznan University of Technology, Poznan, PL

2013 NetFPGA camp, Stanford University, Stanford, US

2010 NetFPGA camp, University of Cambridge, Cambridge, UK

Invited Talks

2021 **Dagstuhl Seminar on Compute-First Networking**, "Network Telemetry on a Budget".

2021 **Workshop on Programmable data plane: Abstractions, Architectures and Applications (co-located with IEEE DRCN)**, "Network Telemetry on a Budget".

2021 **Purdue University (CS536)**, "The State-of-the-art in End-host Centric Networking".

2020 **Huawei Research Germany**, "Network Monitoring in the Age of Programmability".

2020 **Harvard University**, "End-host Networking as first class citizen".

2020 **Zhejiang University**, "Rethinking Networks and Applications with Programmable Data Planes".

2020 **Next Generation Networking UK**, "Network Telemetry on a Budget".

2020 **Italian Networking Workshop**, "Rethinking Networks and Applications with Programmable Data Planes".

2019 **ACM SIGCOMM Topic Preview Session**, "NICs and Switches".

2019 **New York University**, "Dataplane Programming for Fun and Profit".

2018 **University Of Cambridge**, "Enabling Event Triggered Monitoring of Traffic Clusters".

2018 **VMware Research US**, "Understanding PCIe performance for end host networking".

2018 **Microsoft Research Cambridge**, "OSNT and the fine art of art of network system characterisation".

2017 **Lipari PhD Summer School**, "SDN at IXPs: practical challenges".

2017 **ARM Cambridge**, "Networking Systems Characterization in Open Source".

2014 **Doctoral Workshop on Mathematical and Engineering Methods in Computer Science (CZ)**, "Hardware Accelerated Networking Systems: Practice against Theory".

2013 **DELL Inc. US**, "OSNT (Open Source Network Tester)".

2011 **Brno University of Technology**, "Enhancing OpenFlow Capabilities through Pattern Matching Techniques".

2011 **University of Cambridge**, "A Heuristic and Hybrid Hash-Based Approach to Fast Lookup".

PhD students supervision

2020–Present **Alireza Sanaee**, Queen Mary University of London.

2020–Present **Jonatan Langlet**, Queen Mary University of London.

2017–Present **Salvator Galea**, University of Cambridge.

Publications

Journals.

- Fast ReRoute on Programmable Switches, Chiesa M., Sedar R., Antichi G., Borokhovich M., Kamisinski A., Nikolaidis G., Schmid S., **IEEE/ACM Transactions on Networking**, 2021
- An incrementally-deployable P4-enabled architecture for network-wide heavy-hitter detection, Ding D., Savi M., Antichi G., Siracusa D., **IEEE Transactions on Network and Service Management**, 2020
- OFLOPS-SUME and the art of switch characterization, Oudin R., Antichi G., Rotsos C., Moore A.W., Uhlig S. **IEEE Journal on Selected Area of Communication**, Special issue on Scalability Issues and Solutions for Software Defined Networks, 2019
- Rethinking IXPs' Architecture in the Age of SDN, Bruyere M., Antichi G., Fernandes E., Lapeyrade R., Uhlig S., Owezarski P., Moore A.W., Castro I. **IEEE Journal on Selected Area of Communication**, Special issue on Scalability Issues and Solutions for Software Defined Networks, 2019
- ENDEAVOUR: A Scalable SDN Architecture for Real-World IXPs, Antichi G., Castro I., Chiesa M., Fernandes E., Lapeyrade R., Kopp D., Han J., Bruyere M., Dietzel C., Gusat M., Moore A., Owezarski P., Uhlig S., Canini M., **IEEE Journal on Selected Area od Communication**, Special issue on Emerging Technologies in Software-driven Communication, 2017

- Inter-Domain Networking Innovation on Steroids: Empowering IXPs with SDN Capabilities, Chiesa M., Dietzel C., Antichi.G., Bruyere M., Castro I., Gusat M., King T., Moore A.W., Nguyen T.D., Owezarski P., Uhlig S., Canini M., **IEEE Communications Magazine**, Special issue on SDN Use Cases for Service Provider Networks, 2016
- An Open-Source Hardware Approach for High Performance Low-Cost QoS Monitoring of VoIP Traffic, Antichi G., Donatini L., Garroppo R.G., Giordano S., Moore A.W., **Springer Lecture Notes in Computer Science**, 2015
- OSNT: Open Source Network Tester, Antichi G., Shahbaz M., Geng Y., Zilberman N., Covington A., Bruyere M., Feamster N., McKeown N., Felderman B., Blott M., Moore A.W., Owezarski P., **IEEE Network Magazine**, Special issue on Open Source for Networking: Tools and Applications, 2014
- Differential Encoding of DFAs for Fast Regular Expression Matching, Ficara D., Di Pietro A., Giordano S., Procissi G., Vitucci F., Antichi G., **IEEE/ACM Transactions on Networking**, 2011
- A Cooperative PC/Network.Processor Platform for Gigabit Traffic Generation, Antichi G., Di Pietro A.,Ficara D., Giordano S., Procissi G., Vitucci F., **Inderscience International Journal of Communication Networks and Distributed Systems**, Special issue on Performance Evaluation of Computer and Communication Systems, 2009
- Counting Bloom filters for Pattern Matching and Anti-Evasion at the Wire Speed, Antichi G., Ficara D., Giordano S., Procissi G., Vitucci F., **IEEE Network Magazine**, Special issue on Network Intrusion Detection, 2009
- An Improved DFA for fast regular expression matching, Ficara D., Giordano S., Procissi G., Vitucci F., Antichi G., Di Pietro A., **ACM Computer Communication Review**, 2008

Conferences & Workshops.

- Revisiting the Open vSwitch Dataplane Ten Years Later, Tu W., Wei Y., Antichi G., Pfaff B., **ACM SIGCOMM**, Virtual Event, 23-27 August, 2021
- Providing In-network Support to Coflow Scheduling, Benet C.H., Kassler A.J., Antichi G., Benson T., Pongracz G. **IEEE NetSoft**, Tokyo, JP, 28 June-2 July, 2021
- Detecting Routing Loops in the Data Plane, Kucera J., Ben Basat R., Kuka M., Antichi G., Yu M., Mitzenmacher M. **ACM CoNEXT**, Barcelona, ES, 1-4 December, 2020
- Making QUIC quicker with NIC offload, Yang X., Eggert L., Ott J., Uhlig S., Sun Z., Antichi G. **ACM SIGCOMM EPIQ**, New York, New York, US, 10-14 August, 2020
- PINT: Probabilistic In-band Network Telemetry, Ben Basat R., Ramanathan S., Li Y., Antichi G., Yu M., Mitzenmacher M. **ACM SIGCOMM**, New York, New York, US, 10-14 August, 2020
- Full-stack SDN: The Next Big Challenge?, Antichi G., Retvari G. **ACM SOSR**, San Jose, California, US, 3 March, 2020

- Enabling Event-Triggered Data Plane Monitoring, Kucera J., Popescu D., Wang H., Moore A.W., Korenek J., Antichi G. **ACM SOSR**, San Jose, California, US, 3 March, 2020
- Event-Driven Packet Processing, Ibanez S., Antichi G., Brebner G., McKeown N., **ACM HotNets**, Princeton, New Jersey, US, 14-15 November, 2019
- PURR: A Primitive for Reconfigurable Fast Reroute, Chiesa M., Sedar R., Antichi G., Borokhovich M., Kamisinski A., Nikolaidis G., Schmid S., **ACM CoNEXT**, Orlando, Florida, US, 9-12 December, 2019
- An Empirical Study of the Cost of DNS-over-HTTPs, Boettger T., Cuadrado F., Antichi G., Fernandes E., Tyson G., Castro I., Uhlig S., **ACM IMC**, Amsterdam, NL, 21-23 October, 2019
- FAST: Enabling Fast Software/Hardware Prototype for Network Experimentation, Yang X., Sun Z., Li J., Jinli Y., Li T., Quan W., Donglai X., Antichi G., **IEEE/ACM IWQoS**, Phoenix, Arizona, USA, 24-25 June, 2019
- Incremental Deployment of Programmable Switches for Network-wide Heavy-hitter Detection, Ding D., Savi M., Antichi G., Siracusa D., **IEEE NetSoft**, Paris, FR, 24-28 June, 2019
- Understanding PCIe for end host networking, Neugebauer R., Antichi G., Zazo J.F., Audzevich Y., Lopez-Buedo S., Moore A.W., **ACM SIGCOMM**, Budapest, HU, 20-25 August, 2018
- Supporting Emerging Applications With Low-Latency Failover in P4, Sedar R., Borokhovich M., Chiesa M., Antichi G., Schmid S., **ACM NEAT**, Budapest, HU, 20 August, 2018
- An SDN-inspired Model for Faster Network Experimentation, Fernandes E.L., Antichi G., Castro I., Uhlig S., **ACM SIGSIM PADS**, Rome, IT, 23-25 May, 2018
- Re-architecting datacenter networks and stacks for low latency and high performance, Handley M., Raiciu C., Agache A., Voinescu A., Antichi G., Wojcik M., Moore A.W., **ACM SIGCOMM**, Los Angeles, CA, USA, 21-25 August, 2017
- ClassBench-ng: Recasting ClassBench After a Decade of Network Evolution, Matousek J., Antichi G., Lucansky A., Moore A.W., Korenek J., **ACM/IEEE ANCS**, Beijing, CN, 18-19 May, 2017
- Mind the Gap - A Comparison of Software Packet Generators, Emmerich P., GallenmÃijller S., Antichi G., Moore A.W., Carle G., **ACM/IEEE ANCS**, Beijing, CN, 18-19 May, 2017
- Where has my time gone?, Zilberman N., Grosvenor M., Popescu D.A., Manihatty-Bojan N., Antichi G., Wojcik M., Moore A.W., **PAM**, Sidney, AU, 30-31 March, 2017
- OFLOPS-Turbo: Testing the Next-Generation OpenFlow Switch, Rostos C., Antichi G., Bruyere M., Owezarski P., Moore A.W., **IEEE ICC**, London, UK, 8-12 June, 2015
- Towards an SDN Network Control Application for Differentiated Traffic Routing, Adami D., Antichi G., Garroppo R.G., Giordano S., Moore A.W., **IEEE ICC**, London, UK, 8-12 June, 2015

- Blueswitch: Enabling provably consistent configuration of network switches, Han J.H., Mundkur P., Rotsos C., Antichi G., Dave N., Moore A.W., Neumann P.G., **ACM/IEEE ANCS**, Oakland, CA, USA, 7-8 May, 2015
- JA-trie: Entropy-Based Packet Classification, Antichi G., Callegari C., Moore A.W., Giordano S., Anastasi E., **IEEE HPSR**, Vancouver, Canada, 1-4 July, 2014
- On Virtualization-aware Traffic Engineering in OpenFlow Data Centers networks, Gharbaoui M., Martini B., Adami D., Antichi G., Giordano S., Castoldi P., **IEEE/IFIP NOMS**, Krakow, Poland, 5-9 May, 2014
- Datacenter in a Box: Test your SDN Cloud-Datacenter Controller at Home, Antichi G., Adami D., Teixeira J., Del Chiaro A., Giordano S., Santos A., **IEEE EWSDN**, Berlin, DE, 10-11 October, 2013
- From 1G to 10G: Code Reuse in Action, Antichi G., Shahbaz M., Giordano S., Moore A.W., **ACM HPPN**, New York, NY, USA, 18 June, 2013
- Implementation of TCP Large Receive Offload on Open Hardware Platform, Antichi G., Callegari C., Giordano S., **ACM HPPN**, New York, NY, USA, 18 June, 2013
- Effective Resource Control Strategies using OpenFlow in Cloud Data Center, Adami D., Martini B., Antichi G., Giordano S., Gharbaoui M., Castoldi P., **IFIP/IEEE IM**, Ghent, Belgium, 27-31 May, 2013
- An open hardware implementation of CUSUM based Network Anomaly Detection, Antichi G., Callegari C., Giordano S., **IEEE GLOBECOM**, Anaheim, California, USA, 3-7 December, 2012
- Enabling Open-Source High Speed Network Monitoring on NetFPGA, Antichi G., Miller D., Moore A.W., Giordano S., **IEEE/IFIP NOMS**, MAUI, Hawaii, USA, 16-20 April, 2012
- Design and Development of an OpenFlow Compliant Smart Gigabit Switch, Antichi G., Di Pietro A., Giordano S., Procissi G., Ficara D., **IEEE GLOBECOM**, Houston, Texas, USA, 5-9 December, 2011
- Scaling Regular Expression Matching Performance in Parallel Systems through Sampling Techniques, Ficara D., Antichi G., Bonelli N., Di Pietro A., Giordano S., Procissi G., Vitucci F., **IEEE GLOBECOM**, Houston, Texas, USA, 5-9 December, 2011
- On the Use of Compressed DFAs for Packet Classification on NetFPGA, Antichi G., Di Pietro A., Ficara D., Giordano S., Procissi G., Vitucci F., **IEEE CAMAD**, Miami, USA, 3-4 December, 2010
- A Randomized Scheme for IP Lookup at Wire Speed on NetFPGA, Antichi G., Di Pietro A., Ficara D., Giordano S., Procissi G., Vitucci F., **IEEE ICC**, Cape Town, South Africa, 23-27 May, 2010
- Achieving Perfect Hashing through and Improved Construction of Bloom Filters, Antichi G., Di Pietro A., Ficara D., Giordano S., Russo F., Vitucci F., **IEEE ICC**, Cape Town, South Africa, 23-27 May, 2010
- Sampling Techniques to Accelerate Pattern Matching in Network Intrusion Detection Systems, Antichi G., Di Pietro A., Ficara D., Giordano S., Procissi G., Vitucci F., **IEEE ICC**, Cape Town, South Africa, 23-27 May, 2010

- Second-Order Differential Encoding of Deterministic Finite Automata, Antichi G., Di Pietro A., Ficara D., Giordano S., Procissi G., Vitucci F., **IEEE GLOBECOM**, Honolulu, Hawaii, USA, 30 November-4 December, 2009
- Network Topology Discovery through Self-Constrained Decisions, Antichi G., Di Pietro A., Ficara D., Giordano S., Procissi G., Vitucci F., **IEEE GLOBECOM**, Honolulu, Hawaii, USA, 30 Novemeber-4 December, 2009
- A Prefix-Distribution Adaptive Scheme For Routing Lookup Acceleration, Antichi G., Di Pietro A., Ficara D., Giordano S., Procissi G., Vitucci F., **IEEE GLOBECOM**, Honolulu, Hawaii, USA, 30 November-4 December, 2009
- End-to-End Inference of Link Level Queueing Delay Statistics, Antichi G., Di Pietro A., Ficara D., Giordano S., Procissi G., Vitucci F., **IEEE GLOBECOM**, Honolulu, Hawaii, USA, 30 November-4 December, 2009
- A High Level Development, Modeling and Simulation Methodology for Complex Multicore Network Processors, Antichi G., Callegari C., Di Pietro A., Ficara D., Giordano S., Vitucci F., Meneghin M., Torquati M., Vanneschi M., Coppola M., **SPECTS**, Instanbul, Turkey, 13-16 July, 2009
- A Heuristic and Hybrid Hash-based Approach to Fast Lookup, Antichi G., Di Pietro A.,Ficara D., Giordano S., Procissi G., Vitucci F., **IEEE HPSR**, Paris, France, 22-24 June, 2009
- Faster DFAs Through Simple and Efficient Inverse Homomorphisms, Ficara D., Giordano S., Procissi G., Vitucci F., Antichi G., Di Pietro A., **IEEE INFOCOM**, Rio de Janeiro, Brazil, 19-25 April, 2009
- Blooming Trees for Minimal Perfect Hashing, Antichi G., Ficara D., Giordano S., Procissi G., Vitucci F., **IEEE GLOBECOM**, New Orleans, LA, USA, 30 November-4 December, 2008
- Design of a High Performance Traffic Generator on Network Processor, Antichi G., Di Pietro A., Ficara D., Giordano S., Procissi G., Vitucci F., **IEEE DSD**, Parma, Italy, 3 September
- BRUNO: A High Performance Traffic Generator for Network Processor, Antichi G., Di Pietro A., Ficara D., Giordano S., Procissi G., Vitucci F., **SPECTS**, Edimburgh, UK, 16-18 June, 2008